

CNN Reflection

A Convolutional Neural Network, or CNN, is a type of neural network that is commonly used for image data because it can learn patterns from small local regions. Instead of looking at all input values at once, a CNN looks at nearby values together and learns patterns step by step. Early layers learn simple patterns, and deeper layers combine them into more complex features.

A convolution layer works by sliding a small filter across the input and measuring how well each part matches a pattern. The result is a feature map that shows where that pattern appears. After this, an activation function such as ReLU is applied to introduce nonlinearity, which allows the model to learn more complex relationships. Pooling layers are then used to reduce the size of the data and keep the most important information. A typical CNN structure follows a pattern of convolution, activation, pooling, and then fully connected layers for the final prediction.

CNNs are most useful when the data has some form of local structure, such as images or signals. However, they can also be applied to one dimensional inputs, where they still learn patterns between nearby features. This is the approach used in this project.

Connection to the Project

The StatLine project predicts NBA game outcomes using pre game features such as shooting percentages, rebounds, assists, turnovers, recent performance, and team strength using Elo ratings. The original model used in the project is an SVM, which serves as a strong baseline.

For this lab, a 1D CNN was implemented using the same features and compared directly against the SVM. Instead of using a random split, a chronological split was used so that the model trains on earlier games and is tested on later games. This better reflects how predictions would work in real life.

On this evaluation, the SVM achieved about 71.6 percent accuracy and an F1 score of 0.748. The CNN achieved about 73.8 percent accuracy and an F1 score of 0.769, along with slightly higher precision and recall. This shows that the CNN performed better overall on this test.

===== FINAL MODEL COMPARISON =====				
model	accuracy	precision	recall	f1
SVM	0.715556	0.791667	0.708955	0.748031
CNN_1D	0.737778	0.809917	0.731343	0.768627

Even though CNNs are usually used for images, they can still work well in this setting. The features used in this project are already strong and include information about team performance, recent trends, and overall strength. The SVM uses these features to separate winning and losing teams, but it mostly creates a fixed boundary between them.

The CNN is able to go a step further by learning more flexible relationships between the features. Instead of treating each feature independently, it can learn how combinations of features work together. For example, it can learn patterns like how shooting efficiency and recent momentum together affect the outcome of a game.

Because of this, the CNN is able to make slightly better predictions, especially in more complex or close matchups. However, the improvement is not very large, which shows that the SVM is still a strong and reliable model.

Learning Gaps

There are still some areas that need improvement. It is not always clear when a CNN is the best choice for tabular data like this, since the data does not have natural spatial structure. There is also uncertainty about how to choose the number of layers, filters, and other parameters in a CNN. Another area to improve is understanding how different training choices such as learning rate, batch size, and regularization affect the final performance.

Final Takeaway

This lab shows that CNNs can still perform well even when the data is not image based. In this project, the CNN slightly outperformed the SVM on a realistic chronological test, which suggests that it can capture more complex relationships in the data. At the same time, the SVM remains a strong baseline, and the difference between the models is not large. The main takeaway is that model choice matters, but the quality of the features is just as important.